FIITJEE

ALL INDIA TEST SERIES

FULL TEST –XI

JEE (Advanced)-2020

PAPER – 2

TEST DATE: 23-09-2020

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section-A (01 – 08, 19 – 26, 37 – 44)*** *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Full Marks : **+4** If only (all) the four option(s) is (are) chosen.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–1** In all other cases.*

   Section-A (09 – 12, 27 – 30, 45 – 48) *contains 6 List-Match sets with 12 questions (each set has 2 questions). Each question has 4 statements in LIST-I & 4, 5 or 6 statements in LIST-II. The codes for Lists have choices (A), (B), (C), (D) out of which only one is correct.*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

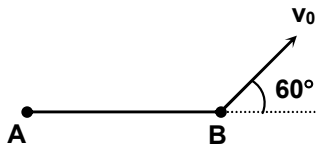
Physics

PART – I

SECTION – A
(One or More than one correct type)

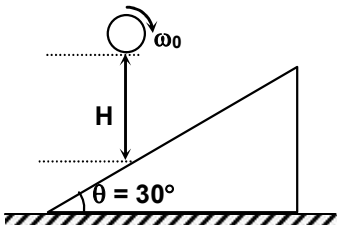
This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. Two identical small balls (mass m each) are connected to each other with the help of an inextensible massless string of length ℓ. The whole arrangement is placed on a horizontal frictionless table with the string just slack. Now ball B is given a sudden horizontal velocity v_0 as shown in the figure Then (m = 1 kg, ℓ = 1 m and v_0 = 1 m/s) just after string becomes taut:



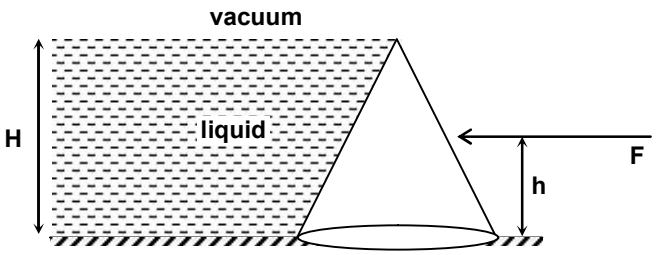
(A) Tension in the string is 0.25 N.
(B) Tension in the string is 0.75 N
(C) Tension in the string is 0.125 N
(D) Relative acceleration between A and B has magnitude 0.75 m/s^2.

2. A ring (mass m and radius r) is rotating with angular speed ω_0 in a vertical plane and is released on an inclined plane of mass M. Friction exists only between ring and wedge (coefficient = μ). It is observed that just after collision, everything is momentarily at rest. Which of the following set(s) of values can satisfy the above observation:



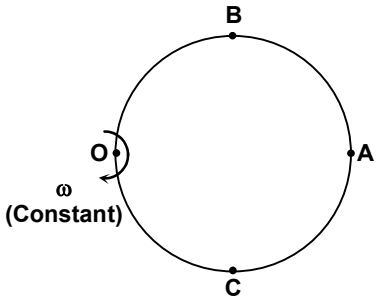
(A) M = 2kg, $\mu = \frac{1}{\sqrt{3}}$, r = 50 cm, H = 5 cm, ω_0 = 1 rad/s, m = 1 kg
(B) M = 1kg, $\mu = \sqrt{3}$, r = 50 cm, H = 10 cm, ω_0 = 1 rad/s, m = 0.5 kg
(C) M = 1kg, $\mu = \frac{1}{2}$, r = 100 cm, H = 20 cm, ω_0 = 2 rad/s, m = 1 kg
(D) M = 1kg, $\mu = \frac{1}{\sqrt{3}}$, r = 100 cm, H = 10 cm, $\omega_0 = \frac{1}{\sqrt{2}}$ rad/s, m = 0.5 kg

3. Imagine a uniform hollow cone of mass M and radius R whose left half is surrounded by a liquid (density ρ). The cone is on a smooth horizontal surface and is kept in equilibrium by applying a particular horizontal force F at a particular height h from ground as shown. Then:



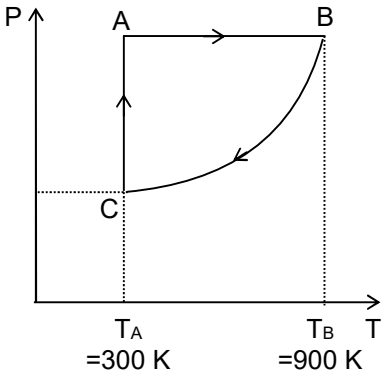
(A) $F = \frac{1}{3} g\rho RH^2$
(B) $h = \frac{H}{4}$
(C) $h = \frac{H}{5}$
(D) $h = \frac{H}{3}$

4. Suppose a uniform steel ring (mass M and radius R) is kept on a smooth horizontal surface with its plane horizontal. It is hinged about a point O on the perimeter & is made to rotate in a horizontal plane as shown (A, B, & C are points on the ring). Then, regarding longitudinal stress developed in the ring:



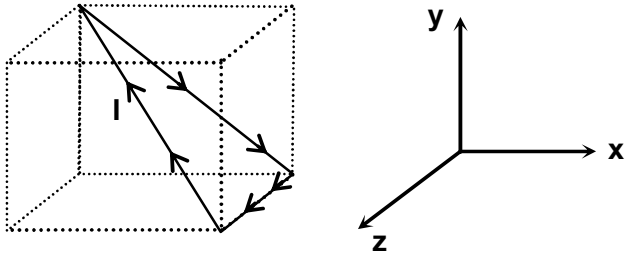
(A) It will be maximum at A
(B) It will be minimum at A
(C) It will be equal at B and C and that value will be maximum.
(D) It will be more at O than A by an amount $\frac{M\omega^2}{\pi^2 R}$.

5. Two moles of an ideal monoatomic gas is taken through a cyclic process ABCA as shown in the P-T diagram. In the process BC, $P = \alpha T^2$, where α is a constant. Then choose the correct option(s). (R = ideal gas constant) [Take $\ell n(3) = 1.1$]



(A) The heat absorbed by the gas during the process AB is 3000 R.
(B) The heat absorbed by the gas during the process BC is 1200 R.
(C) The heat rejected by the gas during the process BC is 600 R.
(D) The efficiency of the cycle is 36%.

6. A current carrying loop is in the form of a triangle whose geometry is as follows:



Dotted (imaginary) lines form a cube of side a. I: current

Then, regarding this loop :

(A) Magnetic field at centre of cube is $-\frac{6\mu_0 i}{4\sqrt{3}\pi a}(\hat{i}+\hat{j})$
(B) Magnetic field at centre of cube is $-\frac{\sqrt{3}\mu_0 i}{6\pi a}(\hat{i}+\hat{j})$
(C) Magnetic field at centre of cube is $+\frac{\sqrt{3}\mu_0 i}{6\pi a}(\hat{i}+\hat{j})$
(D) Magnetic moment of the loop has no z-component.

7. A positive charge (having charge q_0)is thrown from orgin with velocity $v_0(\hat{j}+\hat{k})$ in a gravity-free space. Everywhere in space, there exists a uniform & constant magnetic field $\vec{B}=B_0\,\hat{k}$. Two infinite, rigid, perfectly elastic planar walls are kept at $z = \pm z_0$. Taking mass of the particle to be m_0 & assuming $v_0 > 0$, $z_0 > 0$, choose the correct option(s) :

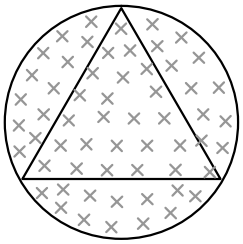
(A) The motion is periodic with period $\frac{4z_0}{v_0}$, for any value of z_0.

(B) The motion is periodic with period $\frac{4z_0}{v_0}$, if z_0 is of the form of $n\frac{2\pi m_0 v_0}{B_0 q_0}$, $n \in I^+$

(C) The motion is periodic with period $\frac{4z_0}{v_0}$, if z_0 is of the form of $\left(\frac{2n-1}{4}\right)\frac{2\pi m_0 v_0}{B_0 q_0}$,, $n \in I^+$

(D) The motion is periodic & the particle crosses origin only once every cycle, if z_0 is of the form $\left(\frac{2n-1}{4}\right)\frac{2\pi m_0 v_0}{B_0 q_0}$, $n \in I^+$

8. In space, there exists a uniform magnetic field in an infinitely long cylindrical region of radius R. The field is given by $\vec{B}=B_0 t\otimes$, where $B_0 > 0$. An insulating charged rigid loop in the shape of an equilateral triangle carries a uniform linear charge density λ and is initially at rest (t = 0). This triangle inscribes the boundary of cylindrical region as shown:



This whole arrangement is in a gravity –free space & take the mass of the triangular loop to be M_0. Then (initially), regarding the triangular loop:

(A) There is no acceleration of centre of mass

(B) There is no angular acceleration of centre of mass

(C) There is angular acceleration = $\frac{3\sqrt{3}}{2}\frac{\lambda B_0 R}{M_0}$ anticlockwise

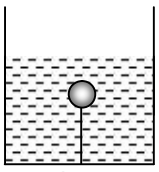
(D) There is angular acceleration = $\frac{9\sqrt{3}}{2}\frac{\lambda B_0 R}{M_0}$ anticlockwise

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

A solid spherical ball is attached to a rigid rod radially and this arrangment is kept inside a liquid (density twice the ball) as shown (with the rod fixed to the container at the bottom):



Now the container is given possible accelerations as in List-I. Some numbers are given in List-II.

	List-I		List-II
(I)	$\vec{a} = \vec{0}$	(P)	2
(II)	$\vec{a} = \frac{g}{2}\downarrow$	(Q)	1
(III)	$\vec{a} = g\downarrow$	(R)	0
(IV)	$\vec{a} = g\uparrow$	(S)	$\frac{3}{2}$
		(T)	$\frac{1}{2}$
		(U)	3

9. If the experiment is done on earth (acceleration due to gravity : g), then match the ratio $\frac{\text{tension (magnitude) in rod}}{\text{weight (magnitude) of ball on earth}}$. Here, weight of ball on earth means mass x acceleration due to gravity.

Options

(A) I $\rightarrow$ S, II $\rightarrow$ Q, III $\rightarrow$ P, IV $\rightarrow$ T
(B) I $\rightarrow$ Q, II $\rightarrow$ T, III $\rightarrow$ R, IV $\rightarrow$ P
(C) I $\rightarrow$ T, II $\rightarrow$ P, III $\rightarrow$ R, IV $\rightarrow$ S
(D) I $\rightarrow$ T, II $\rightarrow$ R, III $\rightarrow$ S, IV $\rightarrow$ P

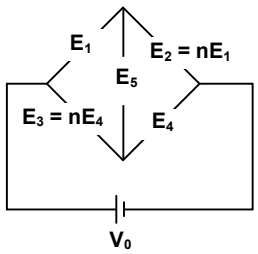
10. Imagine now that the experiment is repeated on a different planet whose density is same as that of earth but radius is twice as that of earth. Then match the ratio $\frac{\text{tension (magnitude) in rod}}{\text{weight (magnitude) of ball on earth}}$.

Options

(A) I $\rightarrow$ T, II $\rightarrow$ U, III $\rightarrow$ P, IV $\rightarrow$ S
(B) I $\rightarrow$ P, II $\rightarrow$ S, III $\rightarrow$ Q, IV $\rightarrow$ U
(C) I $\rightarrow$ Q, II $\rightarrow$ P, III $\rightarrow$ S, IV $\rightarrow$ T
(D) I $\rightarrow$ T, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S

Answer the following by appropriately matching the list based on the information given in the paragraph

E_1, E_2, E_3, E_4, E_5 are five circuit elements such that E_1, E_2, E_3 and E_4 are of same nature (resistive or capacitive) and corresponding E_5 is capacitive or resistive respectively for a given experiment. Four value of n(= 1, 2, 3, 4) give rise to four possible circuits: Circuit-1 (n = 1), circuit -2(n = 2), circuit -3(n = 3) and circuit -4 (n = 4). List I gives the possible four circuit and List II gives some numbers.



n: positive integer

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	List-I		List-II
(I)	Circuit-1	(P)	$\frac{4}{3}$
(II)	Circuit-2	(Q)	$\frac{3}{2}$
(III)	Circuit-3	(R)	1
(IV)	Circuit-4	(S)	$\frac{5}{2}$
		(T)	$\frac{8}{5}$
		(U)	2

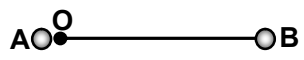
11. If $E_1 = E_4 = R_0$ (resistive) and $E_5 = C_0$(capacitive), then the correct match for the time constant of the voltage acorss the capacitor, as a factor of R_0C_0, is:
Options
(A) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ T, IV $\rightarrow$ P
(B) I $\rightarrow$ S, II $\rightarrow$ P, III $\rightarrow$ R, IV $\rightarrow$ T
(C) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S
(D) I $\rightarrow$ R, II $\rightarrow$ P, III $\rightarrow$ Q, IV $\rightarrow$ T

12. If $E_1 = E_4 = C_0$ (capacitive) and $E_5 = R_0$ (resistive), then the correct match for the time constant of the voltage across the resistor, as a factor of R_0C_0, is:
Options
(A) I $\rightarrow$ R, II $\rightarrow$ Q, III $\rightarrow$ U, IV $\rightarrow$ S
(B) I $\rightarrow$ S, II $\rightarrow$ P, III $\rightarrow$ R, IV $\rightarrow$ U
(C) I $\rightarrow$ R, II $\rightarrow$ P, III $\rightarrow$ Q, IV $\rightarrow$ T
(D) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S

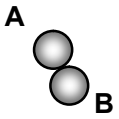
SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. Two identical discs (each of radius r) are arranged such that one is made to fall freely & the other is tied to a string of length ℓ (>>r) as shown:

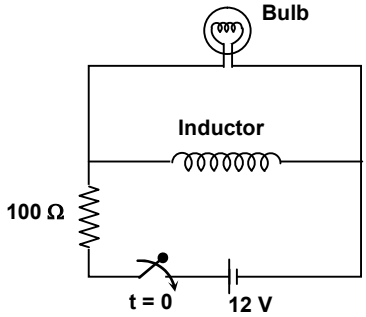


Initially, both discs are at same level (with the string horizontal and disc A just left to the point of suspension (of disc B) O. Now, disc B is released and A is released at such a moment such that following situation arises just before collision (at the lowest point of B):



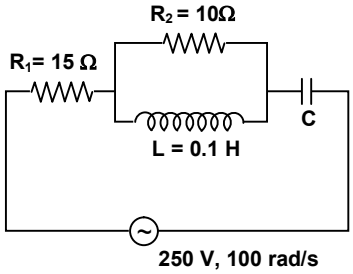
Assuming collision to be perfectly elastic, find the max height in cm, to which B rises (from its lowest position) after the collision. Take $\sqrt{3}$ =1.73. L = 2 m and truncate the final answer to two decimal places.

14. A bulb (purely resistive of resistance 100 Ω) is connected in a circuit using a DC battery of 12 V as shown :



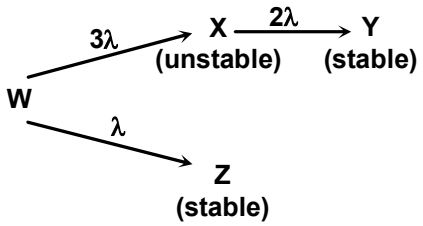
The inductor has an inductance of 0.5 mH & resistance of 100 Ω. The bulb stops glowing if current passing through it becomes any less than 50 mA. Then for what time interval (in μs), after closing the switch, will the bulb glow for. (Take ℓn 2 = 0.69)

15. In the circuit shown, if the current supplied by the ac voltage source is 10 A. Find the value of capacitance 'C' in μF.



16. A container consists of a sample of hydrogen like ions of the same element & in the same state. In this state, the ionization energy of an ion in the above sample is 36% of a hydrogen atom. This sample is now allowed to reach its most stable configuration, thereby resulting in 10 distinct types of photons. The most energetic Paschen – line photons from this sample are made to incident on a metal. The stopping potential of the ejected photoelectrons is observed to be 4V. Find the work function (in eV) of the metal. (Take ionisation energy of a hydrogen atom= 13.6 eV and truncate your final answer to two decimal places)

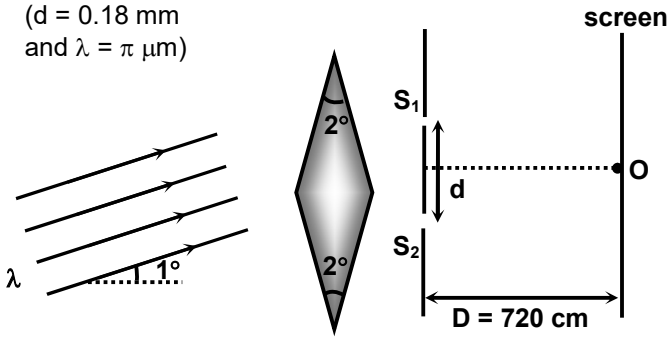
17. A radioactive element W exhibits radioactivity in the following way:



The decay constants have been shown (λ = 2 minutes^{-1}). Initially, only element W was present. Find in how many minutes, will the number of nuclei of element X would equal the number of nuclei of element Z. Take ℓn5 = 1.61 & truncate yours final answer to two decimal places).

18. A very thin biprism is made up of a material of refractive index 1.5 & angle of 2^0 for both upper apex & lower apex. This is used in a YDSE setup such that upper prism is near upper slit & lower prism is near lower slit is as shown:

(d = 0.18 mm and $\lambda = \pi$ μm)



A parallel beam of light (inclined at 1^0 to the horizontal) is incident on the setup as shown. Find (in cm) the separation between the point O (which is the geometrically symmetric point of the YDSE) & the closest point on the screen where intensity is half the maximum possible.
(Take π = 3.14, sin θ = θ = tan θ for small angles)

Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. Consider the following reactions (unbalanced)

$Cu + \text{conc.}HNO_3 \longrightarrow K + L\uparrow + M$

$K + KCN(\text{excess}) \longrightarrow T + S\uparrow + KNO_3$

Choose the correct option(s)

(A) The oxidation state of Cu in T is +2.
(B) 'S' has 4π bonds.
(C) 'L' is brown gas.
(D) 'T' on reaction with H_2S does not give precipitate of CuS.

20. In Baeyer's process (extraction for aluminium metal), bauxite ore is first digested with reagent 'Q' to form 'R'. The solution of 'R' is then neutralized by passing gas 'S' and 'T' is precipitated out. Choose the correct option(s)

(A) 'Q' is NaOH
(B) 'R' is $Na[Al(OH)_4]$
(C) Bauxite ore is $Al(OH)_3$
(D) 'S' is CO_2 gas

21. With reference to Cl_2, choose the correct option(s):

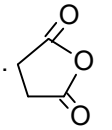
(A) Excess amount Cl_2 when react with NH_3 gives an explosive substance NCl_3.
(B) Cl_2 on reaction with hot and conc. NaOH gives chloride and chlorate.
(C) Bleaching action of Cl_2 is due to reduction.
(D) Cl_2 water convert ferric ion into ferrous ion in acidic medium.

22. Each of the following options contains a set of four molecules. Identify the option(s) where all bond lengths in a molecules are equal in all four molecules in an option

(A) OF_2, SF_4, XeF_2, ClF_3
(B) $NO_3^-, O_3, [SF_5]^+, [ClF_4]^+$
(C) $NO_3^-, BF_3, SiF_4, SH_2$
(D) $XeF_2, P_4, I_3^-, SO_4^{2-}$

23. Choose the correct option(s) for the following reaction sequence:

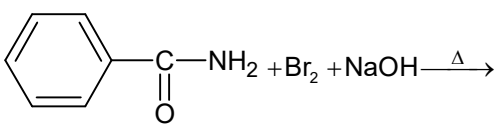
$$H-C\equiv C-H \xrightarrow{\text{1. Red hot Fe tube}} \text{'P'} \xrightarrow[\Delta]{\text{Conc. } H_2SO_4} \text{'Q'}$$

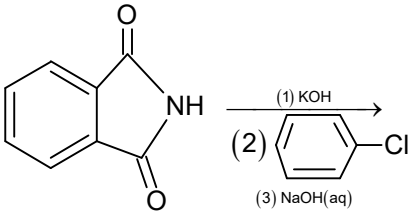
2.  Anh. $AlCl_3, \Delta$

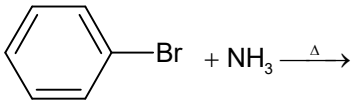
3. $Zn / Hg, HCl, \Delta$

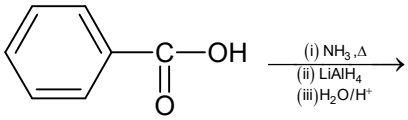
(A) The compound 'P' gives 2,4-DNP test.
(B) The compound 'Q' gives effervescence of CO_2 with $NaHCO_3$.
(C) The conversion of 'P' to 'Q' is a type of electrophilic substitution reaction.
(D) The compound 'P' is soluble in NaOH.

24. Which of the following reaction (s) do not/does not give aniline as major product?

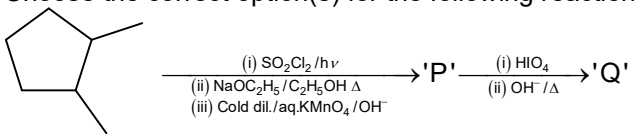
(A) 

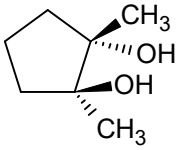
(B) 

(C) 

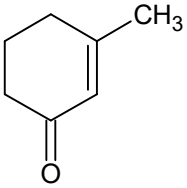
(D) 

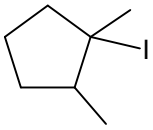
25. Choose the correct option(s) for the following reaction sequence



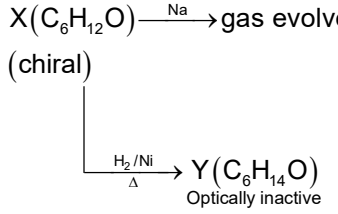
(A) The compound 'P' is 

(B) Total 'P' products formed are two.

(C) The compound 'Q' is 

(D) Compound 'P' on reaction with excess of HI gives 

26. Consider the following reaction:



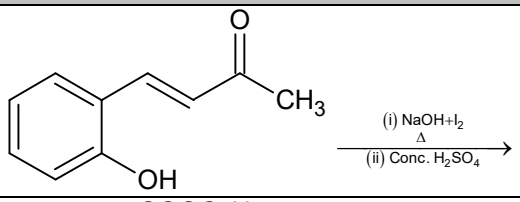
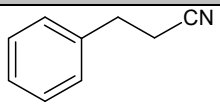
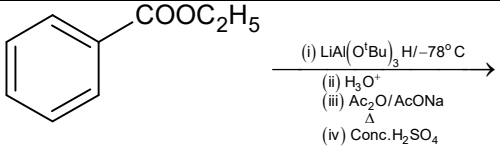
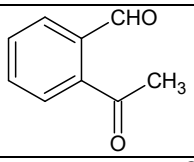
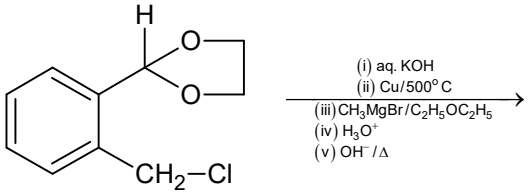
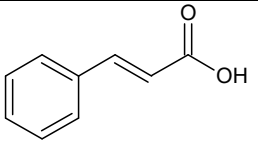
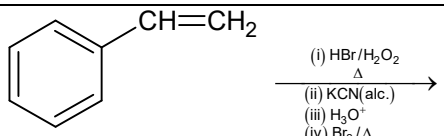
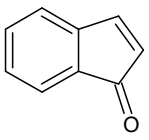
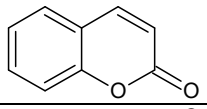
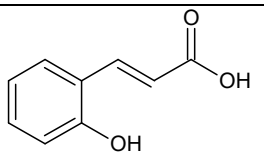
(A) Only 'X' gives yellow solid with $NaOH/I_2$

(B) Both X and Y form immediate turbidity with HCl in the presence of anh. $ZnCl_2$.
(C) Only 'X' decolourises the colour of acidic solution $KMnO_4$.
(D) Dehydrogenation takes place when 'Y' is heated with Cu at 300°C.

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

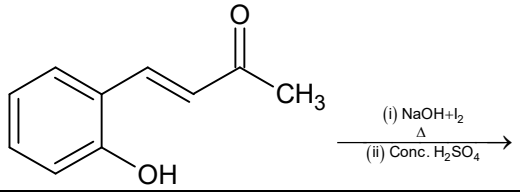
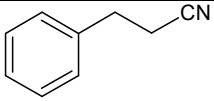
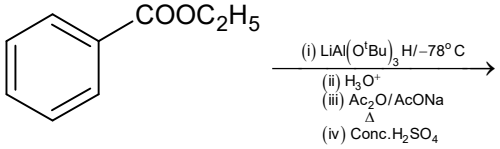
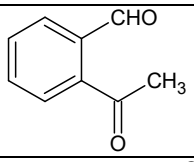
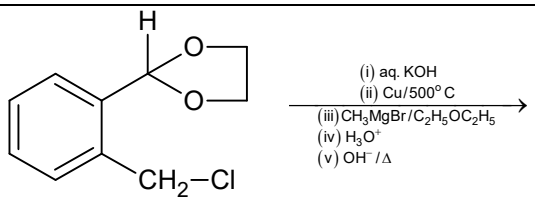
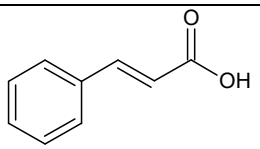
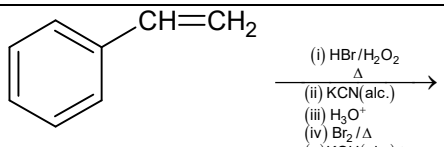
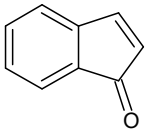
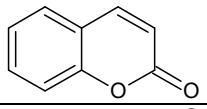
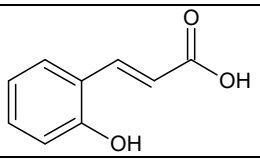
Answer the following by appropriately matching the lists based on the information given in the paragraph.
List-I includes starting materials and reagents of selected chemical reactions. List-II gives structures of compounds that may be formed as intermediate products and/or final products from the reactions of List-I.

LIST–I		LIST–II	
(I)		(P)	
(II)		(Q)	
(III)		(R)	
(IV)		(S)	
		(T)	
		(U)	

27. Which of the following options has correct combination considering List-I and List-II?
(A) (III), (T), (U)
(B) (IV), (R), (T)
(C) (IV), (P), (S)
(D) (III), (Q), (S)

Answer the following by appropriately matching the lists based on the information given in the paragraph.

List-I includes starting materials and reagents of selected chemical reactions. List-II gives structures of compounds that may be formed as intermediate products and/or final products from the reactions of List-I.

LIST–I		LIST–II	
(I)		(P)	
(II)		(Q)	
(III)		(R)	
(IV)		(S)	
		(T)	
		(U)	

28. Which of the following options has correct combination considering List-I and List-II?

(A) (I), (U), (S)
(B) (II), (R), (T)
(C) (I), (U), (T)
(D) (II), (P), (T)

Answer the following by appropriately matching the lists based on the information given in the paragraph.

In the following List-I contains equation of state of a gas in different conditions of pressure and List-II contains properties of that gas (P = pressure of gas, V_m = molar volume of gas, T = temperature of gas, Z = compressibility factor, U_{rms} = root mean square velocity of gas, M = molar mass of gas, a and b are van der Waals constant, R = gas constant)

LIST–I		LIST–II	
(I)	$P(V_m - b) = RT$	(P)	At constant temperature the value of PV_m does not change by changing pressure
(II)	$PV_m = RT - \frac{a}{V_m}$	(Q)	$Z < 1$
(III)	$PV_m = \frac{1}{3}MU_{rms}^2$	(R)	Z is independent of P and T
(IV)	PM = dRT	(S)	Attractive forces are dominant between gas molecules
		(T)	Z increases with increasing P.
		(U)	Z = 0

29. Which of the following options has correct combination considering List-I and List-II?
(A) (II), (P)
(B) (III), (T)
(C) (III), (Q)
(D) (II), (S)

Answer the following by appropriately matching the lists based on the information given in the paragraph.

In the following List-I contains equation of state of a gas in different conditions of pressure and List-II contains properties of that gas (P = pressure of gas, V_m = molar volume of gas, T = temperature of gas, Z = compressibility factor, U_{rms} = root mean square velocity of gas, M = molar mass of gas, a and b are van der Waals constant, R = gas constant)

LIST–I		LIST–II	
(I)	$P(V_m - b) = RT$	(P)	At constant temperature the value of PV_m does not change by changing pressure
(II)	$PV_m = RT - \frac{a}{V_m}$	(Q)	$Z < 1$
(III)	$PV_m = \frac{1}{3}MU_{rms}^2$	(R)	Z is independent of P and T
(IV)	PM = dRT	(S)	Attractive forces are dominant between gas molecules
		(T)	Z increases with increasing P.
		(U)	Z = 0

30. Which of the following options has correct combination considering List-I and List-II?
(A) (IV), (U)
(B) (I), (R)
(C) (I), (T)
(D) (IV), (Q)

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. A current strength of 9.65 A is passed for 10 seconds through 1 L of solution of 0.1 M aqueous $CuSO_4$. The pH of solution is…..(Assuming that during the process there is no change in volume).

32. AB_2 dissociates in A^{2+} and B^- ions in an aqueous solution. What is depression of freezing point of aqueous AB_2 solution containing 0.65 mole fraction of AB_2 in 930 gm of water, if the ratio of the observed depression of freezing point of the aqueous solution to the value of the depression of freezing point in the absence of ionic dissociation is 2. Given K_f of water is 1.86 K kg mol^{-1}.

33. In the sequence of the following:

$$^{238}_{92}X \xrightarrow[(\alpha)]{-(1)^4_2He} Y \xrightarrow[(\beta^+)]{-(2)^0_1e} Z \xrightarrow[(\beta^-)]{-(1)^0_{-1}e} L \xrightarrow[(\alpha)]{-(n)^4_2He} {}^{206}_{(a)}M$$

The value of (n + a) is

34. The ammonia prepared by treating ammonium sulphate with NaOH is used by $CuSO_4.5H_2O$ to form a stable coordination compound. If 330 g of pure ammonium sulphate and 499 g of pure $CuSO_4.5H_2O$ are used in preparation and NaOH is used the excess, the moles of copper – ammonia coordination compound produced is
(Atomic weight in g mol^{-1} H = 1, N = 14, O = 16, S = 32, Na = 23, Cu = 63.5)

35. Consider the following reaction

$$XeF_4 + H_2O \longrightarrow A + B + HF + O_2$$

A and B are Xe containing compounds. The oxidation state of Xe in compound 'A' is x and in compound 'B' is y. Then what is x + y?

36. Based on VSEPR theory, the number of 180° F – Xe – F angles in a molecule of $[XeF_3]^+$ $[SbF_6]^-$ is

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If all the values of x which satisfies the inequality $\log_{1/3}(x^2 + 2Px + P^2 + 1) \geq 0$ also satisfy the inequality $kx^2 - kx - k^2 \leq 0$ for all real values of k, then all possible values of p lies in the interval
(A) [–1, 1]
(B) [0, 1]
(C) [0, 2]
(D) [–2, 0]

38. The real root of the equation $8x^3 - 3x^2 - 3x - 1 = 0$ can be written in the form $\frac{\sqrt[3]{m} + \sqrt[3]{n} + 1}{P}$ where m, n and P are the positive integers ($m \geq n \geq P$), then
(A) m + n + P = 98
(B) m + n + P = 97
(C) m + n = 90
(D) n + P = 17

39. If 5x – y, 2x + y, x + 2y are in A.P. and $(x - 1)^2$, (xy + 1), $(y + 1)^2$ are in G.P. $x \neq 0$, then (x + y) equals
(A) $\frac{3}{4}$
(B) 3
(C) –5
(D) –6

40. If the equation $|z|(z + 1)^8 = z^8|z + 1|$ where z is a complex number and $z(z + 1) \neq 0$ has distinct roots $z_1, z_2, z_3 \ldots z_n$ (where $n \in N$), then
(A) $z_1, z_2, z_3, \ldots z_n$ are cyclic points
(B) $z_1, z_2, z_3, \ldots z_n$ are collinear points
(C) $\sum_{r=1}^{n} Re(z_r) = \frac{-7}{2}$
(D) $\sum_{r=1}^{m} I_m(z_r) = 0$

41. If $^{100}C_{50}$ can be prime factorized as $2^{\alpha} \cdot 3^{\beta} \cdot 5^{\gamma} \cdot 7^{8} \ldots$ where $\alpha, \beta, \gamma, \delta$ are non negative integers, then
(A) $\alpha < \beta$
(B) $\gamma < \delta$
(C) $\alpha + \delta = \beta + \gamma - 1$
(D) $\gamma + \delta = 0$

42. $f(x) = [x^2 - x + 1]$, $x \in (0, 2)$, then f(x) is discontinuous at (where [.] denotes greatest integer function)

(A) $\frac{1+\sqrt{2}}{2}$

(B) $\frac{1+\sqrt{3}}{2}$

(C) $\frac{1+\sqrt{5}}{2}$

(D) 1

43. Let $(x + 1)(x + 2)(x + 3) \ldots\ldots (x + 100) = a_0 + a_1x + a_2x^2 + \ldots\ldots a_{100}x^{100}$, then $100a_0 + 99a_1 + 98a_2 + \ldots\ldots a_{99}$ is equal to

(A) $(101)!\left(\frac{1}{2}+\frac{2}{3}+\ldots..+\frac{100}{101}\right)$

(B) $(101)!\left(1+\frac{1}{2}+\frac{2}{3}+\ldots..+\frac{1}{100}\right)$

(C) $(101)!\left(1+\frac{1}{2}+\frac{2}{3}+\ldots..+\frac{100}{101}\right)$

(D) $(101)!\sum_{n=4}^{103}\left(\frac{n-3}{n-2}\right)$

44. Let $f(x) = \int_0^{x^2} \frac{t^2 - 5t + 4}{2 + e^t}$, then f(x) has

(A) minimum at x = –2, 0, 2

(B) maxima at x = –1, 1

(C) minima at x = –2, 2

(D) maxima at x = –1, 0, 1

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the lists based on the information given in the paragraph.

MM-I. Match the following list(s)

	LIST-I		**LIST-II**
(I)	If $\vec{a} = \hat{j} + \sqrt{3}\hat{k}$, $\vec{b} = \hat{j} + \sqrt{3}\hat{k}$ and $\vec{c} = 2\sqrt{3}\hat{k}$ from a triangle, then the internal angles of the triangle between $\vec{a}$ and $\vec{b}$ is	(P)	$\frac{\pi}{6}$
(II)	If $\int_a^b (f(x) - 3x)dx = a^2 - b^2$, then the value of $f\left(\frac{\pi}{6}\right)$ is	(Q)	$\frac{2\pi}{3}$
(III)	The value of $\frac{\pi^2}{\ln 3}\int_{7/6}^{5/6} \sec(\pi x)dx$ is	(R)	$\frac{\pi}{3}$
(IV)	The maximum value of $\left\lvert \arg\left(\frac{1}{1-z}\right)\right\rvert$ for $\lvert z\rvert = 1$, $z \neq 1$ is approaching to the value	(S)	π
		(T)	$\frac{\pi}{2}$
		(U)	$\frac{\pi}{4}$

45. Which of the following is the only CORRECT combination?
(A) (I), (P)
(B) (II), (P)
(C) (III), (P)
(D) (IV), (Q)

46. Which of the following is the only INCORRECT combination?
(A) (I), (Q)
(B) (III), (S)
(C) (IV), (T)
(D) (III), (T)

Answer the following by appropriately matching the lists based on the information given in the paragraph.

MM-II. Match the following list(s)

LIST-I		LIST-II	
(I)	The set $\left\{\text{Re}\left(\frac{2iz}{1-z^2}\right) z \text{ is a complex number } \lvert z\rvert = 1, z \neq \pm 4\right\}$ is	(P)	$(-\infty, -1) \cup (1, \infty)$
(II)	If $f(\theta) = \begin{vmatrix} 1 & \tan\theta & 1 \\ \tan\theta & 1 & \tan\theta \\ -1 & \tan\theta & 1 \end{vmatrix}$, then the set $\left\{f(\theta), 0 \le \theta < \frac{\pi}{2}\right\}$ is	(Q)	$(-\infty, 0) \cup (0, \infty)$
(III)	The domain of the function $f(x) = \sin^{-1}\left(\frac{8(3)^{x-2}}{1-3^{2(x-1)}}\right)$ is	(R)	$[2, \infty)$
(IV)	If $f(x) = x^{3/2}(3x - 10)$, $x \ge 0$, then $f(x)$ is increasing in	(S)	$(-\infty, -1] \cup [1, \infty)$
		(T)	$(-\infty, 0] \cup [2, \infty)$
		(U)	$(0, \infty)$

47. Which of the following is the only CORRECT combination?
(A) (I), (S)
(B) (II), (T)
(C) (III), (R)
(D) (IV), (S)

48. Which of the following is the only INCORRECT combination?
(A) (II), (R)
(B) (III), (T)
(C) (IV), (R)
(D) (I), (Q)

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. If $a_n = \sqrt{1+\left(1-\frac{1}{n}\right)^2} + \sqrt{1+\left(1+\frac{1}{n}\right)^2}$ $(n \ge 1)$, then the value of $\sum_{i=1}^{20} \frac{1}{a_i}$ is equal to

50. Sum of the coefficients in the expansion of $(2x + y + z)^{37}$ which are divisible by 37 can be written as $(a^p - b^p - c)$ where p is a prime number and a, b, c are even, then $\left(\frac{a+b+c+2}{2}\right)$ is equal to

51. A point P is randomly selected from the region described by $3 \le |z| \le 5$ (where z is a complex number). If the probability that argument of P lies in the interval $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ is $\frac{1}{N}$, then N is

52. Let $f(x) = ax^4 + 6x^2 + 3x + 7$ and $f(-4) = 2286$ and $f(4) = N$. Then the number of ways in which the number N can be resolved as a product of two divisor which are relatively prime

53. If the distance of two points P and Q from the focus of a parabola $y^2 = 4ax$ are 4 and 9 respectively, then the distance of the point of intersection of tangents at P and Q from the focus is

54. Let f(x) be continuous function such that $f(0) = 1$, $f(x) - f\left(\frac{x}{7}\right) = \frac{x}{7}$ $\forall$ $x \in R$. The area of region bounded by the curve $y = f(x)$ and coordinate axes is